
CampusMonk
By Rachit Rastogi

Ques

- **Input**
- The input consist of three space-separated integers- input1, input2, and input3 representing the three four-digit numbers.
- **Output**
- Print an integer representing the concatenation of four digits where first digit represents the smallest digit in the thousands places of all three numbers; second digital represents the largest digital in the hundreds place of all three numbers; third digit represents the smallest digit in the tens place of all three numbers and fourth digit represents the largest digit in the units place of all represents the largest digit in the units place of all three numbers
- **Example**
- Input:
- 3521 2452 1352
- Output:
- 1522
- **Explanation**
- Key = [smallest digit in the thousands places of all three numbers] [LARGEST digit in the hundreds place of all three numbers] [smallest digit in the tens place of all three numbers] [LARGEST digit in the units place of all three numbers]
- If input1 = 3521, Input2 = 2452, Input3 = 1352, then key = [1] [5] [2] [2] = 1522

```
#include<bits/stdc++.h>
using namespace std;
int main()
{
   int a,b,c; cin>>a>>b>>c;
   int ans = 0;
   for(int i=0;i<4;i++)
   {
            if(i%2==0)
                     ans += pow(10,i)*max({a%10, b%10, c%10});
            else
                     ans += pow(10, i)*min({a%10, b%10, c%10});
            a/=10;
            b/=10;
            c/=10;
   }
   cout<<ans;
}
```

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For Python and JAVA try to code it yourself,
if any doubt then ask,
it will help you to become more proficient in that language

Ques

At “YoursPay” bank, the customer bank lockers use a digital lock. The digital lock is encoded in numeric form. To decode the lock there are two methods. If the lock is encoded by an Armstrong number, then the decoding number is the sum of the even digits of the encoded number. If the encoded number is not an Armstrong number, then the decoding number is the sum of the odd digits of the encoded number. An Armstrong number is a number that is the sum of its own digits each raised to the power of the number of digits in the number.

Write an algorithm to decode the digital lock

Input

The Input is a long integer – encodedLock representing the numeric value of the encoded digital lock

Output

Print an integer representing the decoding number for the encoding digital lock.

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```cpp
#include<bits/stdc++.h>
using namespace std;
long long power(long long x, long long n)
{
    long long ans = 1;
    for(int i=0;i<n;i++)
    {
        ans = ans*x;
    }
    return ans;
}
bool isArmstrong(long long n)
{
    long long originalNum = n;
    long long count = 0;
    long long num = n;
    // //no of digits
    long long temp = num;
    while(temp!=0)
    {
        temp/=10;
        count++;
    }
    num = n;
    long long x = 0;
    while(num!=0)
    {
        x += power(num%10, count);
        num/=10;
    }
    if(x == n)
    {
        return true;
    }
    return false;
}
```

```cpp
int main()
{
    long long num;  cin>>num;
    int ans = 0;
    if(isArmstrong(num))
    {
        //even
        while(num!=0)
        {
            int last_digit = (num%10);
            if(last_digit%2==0)
            {
                ans+=num%10;
            }
            num/=10;
        }
    }
    else
    {
        //odd
        while(num!=0)
        {
            int last_digit = (num%10);
            if(last_digit%2!=0)
            {
                ans+=num%10;
            }
            num/=10;
        }
    }
    cout<<ans<<endl;
}
```

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Ques

Given two arrays arr1 and arr2 of size n and m respectively re-arrange elements of arr1 such that the relative order of elements in arr1 is same as order of elements in arr2 and the remaining elements which are in arr1 but not in arr2 should be sorted in ascending order.

example:

input:

n = 9, m = 4

arr1 = [9, 1, 10, 1, 2, 5, 99, 1, 3]

arr2 = [2, 1, 9, 5]

output:

arr1 = [2, 1, 1, 1, 9, 5, 3, 10, 99]

Code in Cpp(c++)

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```cpp
#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,m;    cin>>n>>m;
    vector<int>arr1(n),arr2(m);
    for(int i=0;i<n;i++)
    {
        cin>>arr1[i];
    }
    for(int i=0;i<m;i++)
    {
        cin>>arr2[i];
    }
    map<int, int>mp;
    for(int i=0;i<n;i++) mp[arr1[i]]++;
    int j=0;
    for(int i=0;i<m;i++)
    {
        while(mp[arr2[i]]-- > 0)
        {
            arr1[j++] = arr2[i];
        }
    }
    for(auto &it: mp)
    {
        while(it.second-- > 0)
        {
            arr1[j++] = it.first;
        }
    }
    for(int it: arr1)
    {
        cout<<it<<" ";
    }
}
```

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Homework

Sort elements of an array by frequency of its elements(high frequency to low frequency)

Find non repeating characters of a string/

Find Maximum occurring character in a string

Find the largest word in a string